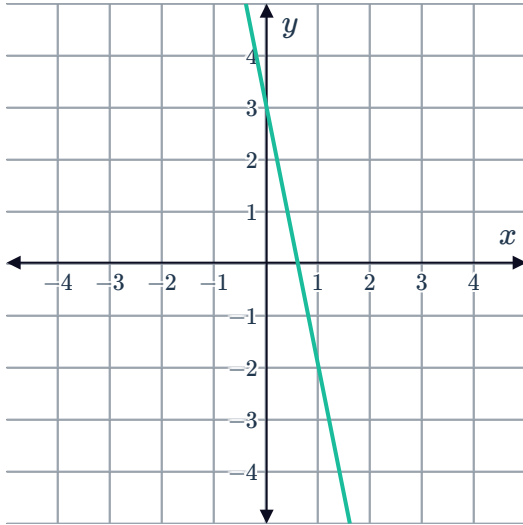


Characteristics from a graph

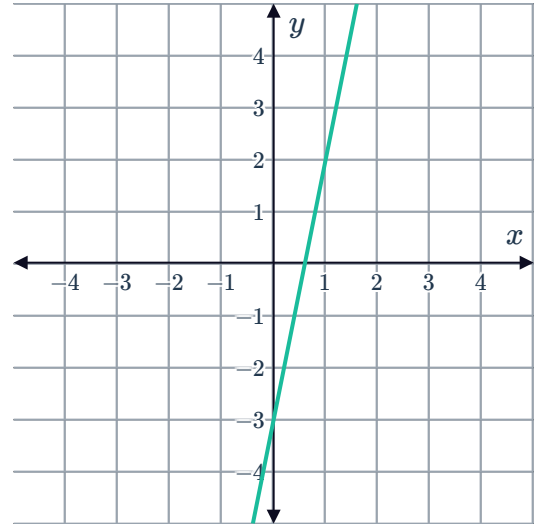


1 State whether each of the following graphs is a linear function decreasing from left to right.

a

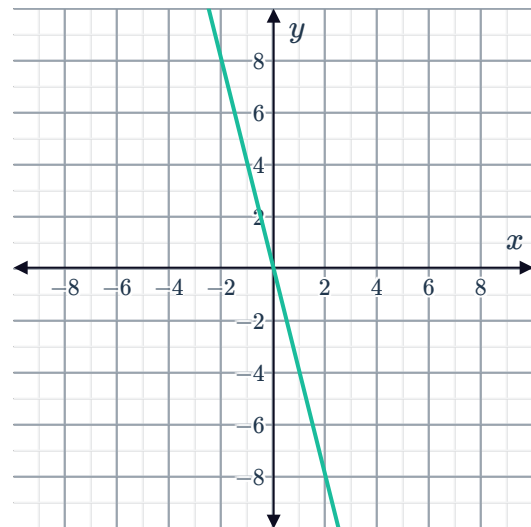


b



2 Consider the graph of $y = -4x$. Describe the end behavior of the y -values:

- a As x becomes larger in the positive direction (i.e. x approaches ∞).
- b As x becomes larger in the negative direction (i.e. x approaches $-\infty$).

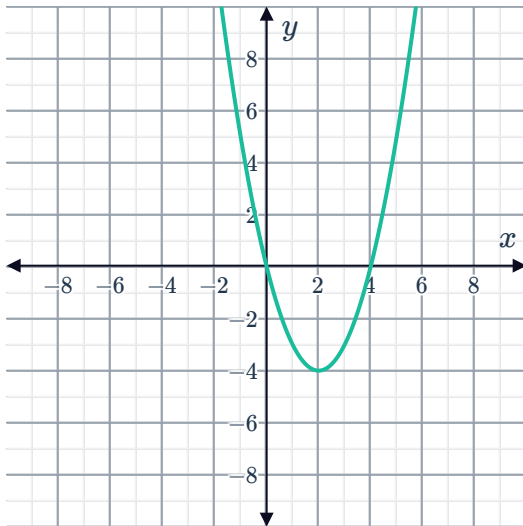


3 Consider the following graphs.

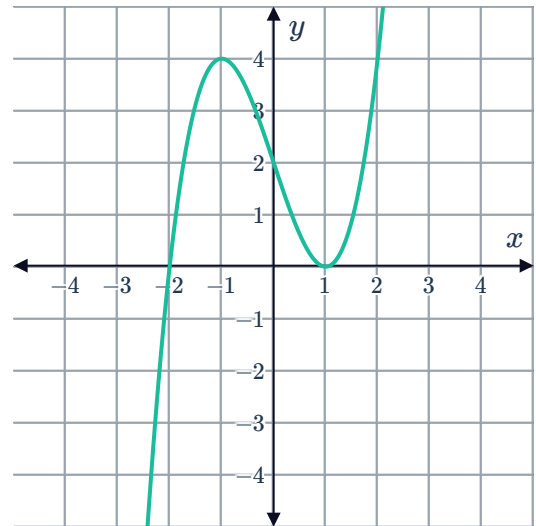


- i State whether the function has an absolute maximum or minimum.
- ii State whether the function has local maximums or minimums.

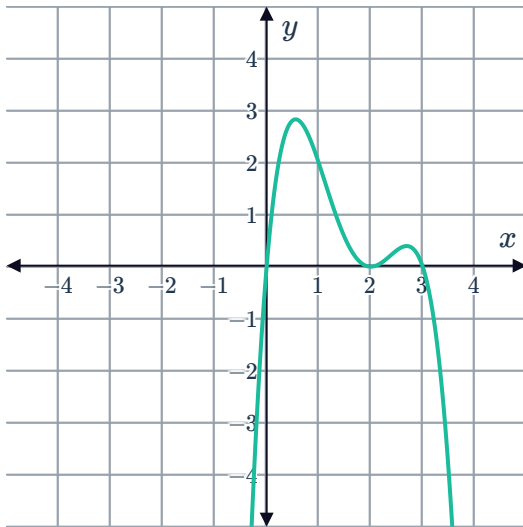
a



b



c

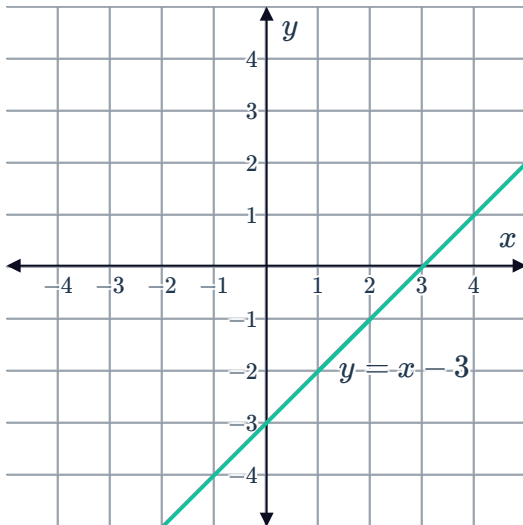


4 Consider the following graphs:

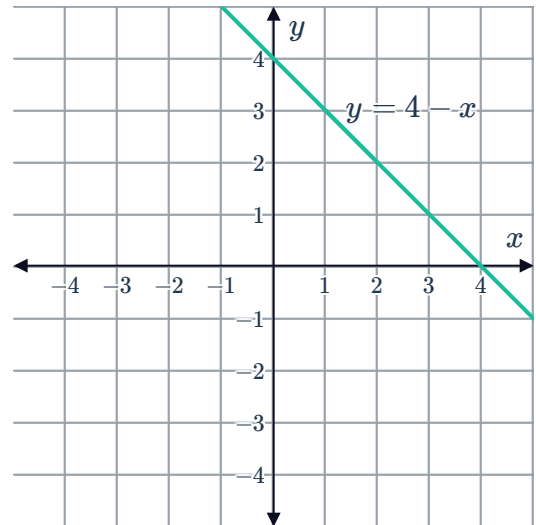


- i State the coordinates of the x -intercept.
- ii State the coordinates of the y -intercept.

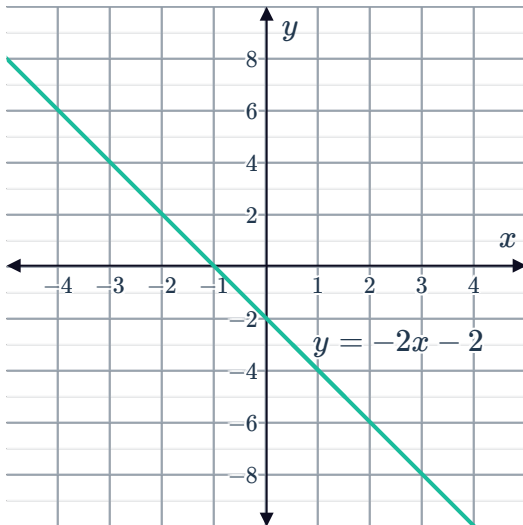
a



b



c

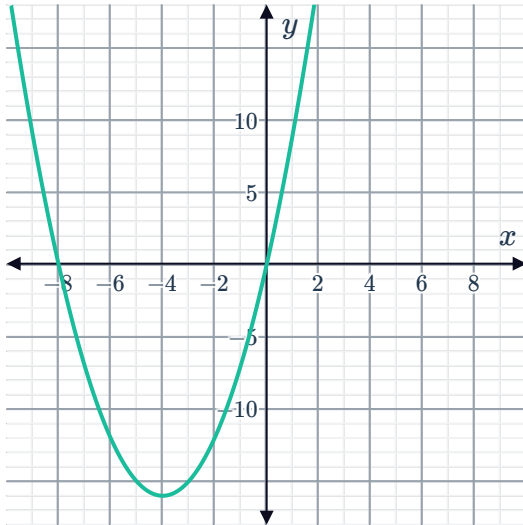


5 Consider the functions shown on the following graphs:

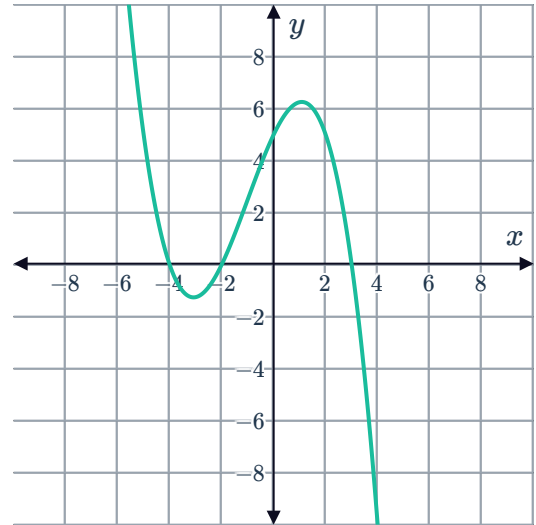


- i State the coordinates of the x -intercepts.
- ii State the coordinates of the y -intercept.
- iii As $x \rightarrow \infty$, describe the end behavior of the corresponding y -values.
- iv As $x \rightarrow -\infty$, describe the end behavior of the corresponding y -values.

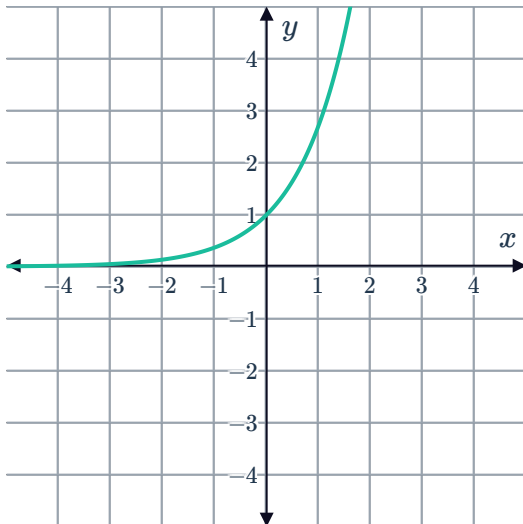
a



b



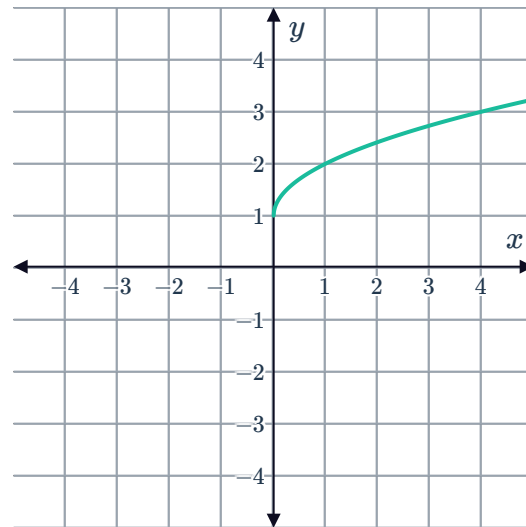
c



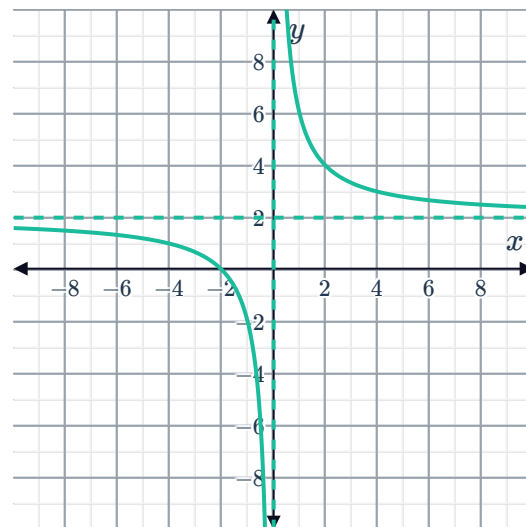
6 Consider the graph below.



- a State the coordinates of the x -intercept.
- b State the coordinates of the y -intercept.
- c As x becomes larger in the positive direction (ie. x approaches ∞), describe the end behavior of the y -values.

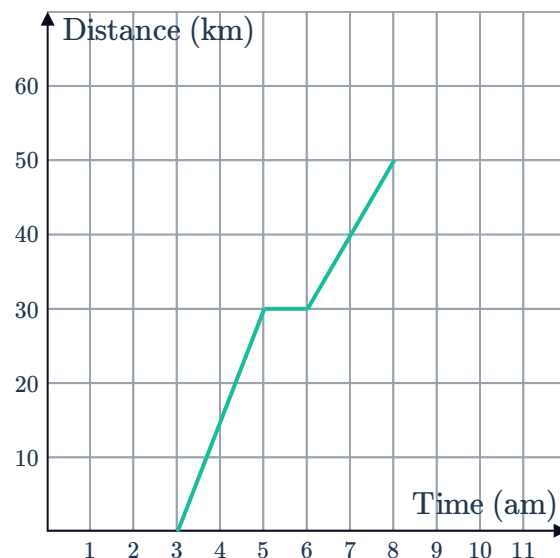


7 State the coordinates of the x -intercept of the following graph:

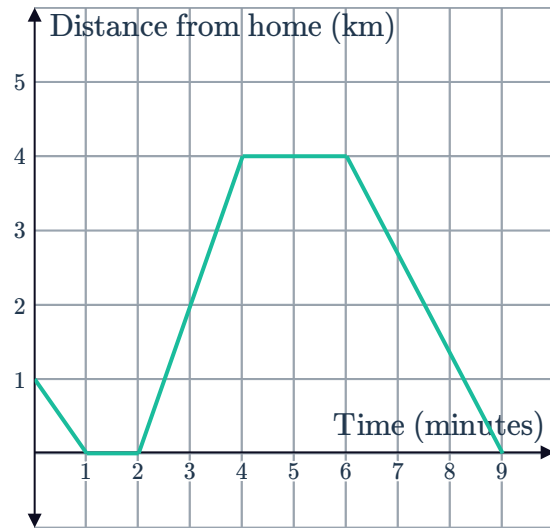


8 The graph shown describes Neil's distance from home throughout his journey.

- a How far was Neil from home when he started driving at 3 am?
- b How far was he from home by 5 am?
- c How long did he take to rest?



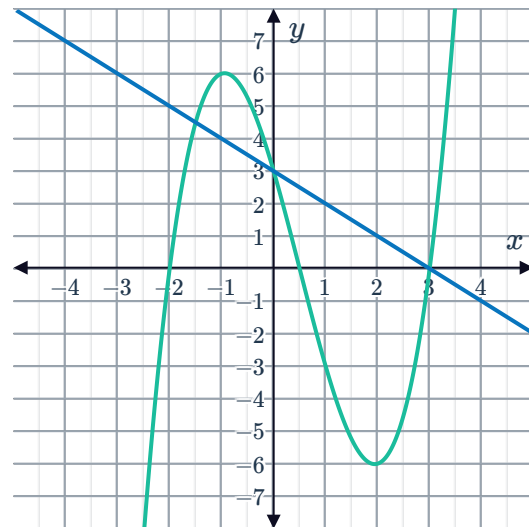
- 9 The graph represents Judy's distance from home over a 9 minute interval. Determine Judy's initial distance from home.



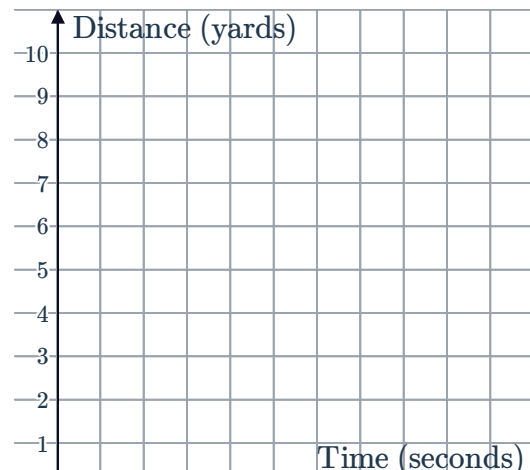
- 10 Consider the function $f(x) = -x^2 + 3x - 2$. Determine the interval for which interval the function is both decreasing and positive.

- 11 Consider the functions $f(x)$ and $g(x)$ shown below. Determine whether $f(x) \neq g(x)$ for each of the following values of x .

- | | |
|------|------------------|
| a 0 | b 3 |
| c -3 | d $-\frac{3}{2}$ |



- 12 A puppy is playing outside. Observe the following graph which shows the puppies distance $d(t)$, in yards from home over a period of time, t , in seconds. Determine the interval representing the puppy constantly moving further from home.





13 The amount of water (million gallons per hour) flowing into a dam is recorded. For the first 4 hours, it flowed at an average rate of 1 million gallons per hour. The water then started to flow at an average rate of 0.5 million gallons per hour for the next 3 hours. Then it stopped flowing for 2 hours. Then it started flowing again at an average rate of 1.5 million gallons per hour for the next 3 hours.

- a** Graph the function showing the accumulated water flow over the recorded period.
- b** Determine the average rate of flow over the length of the recorded period.

Additional questions

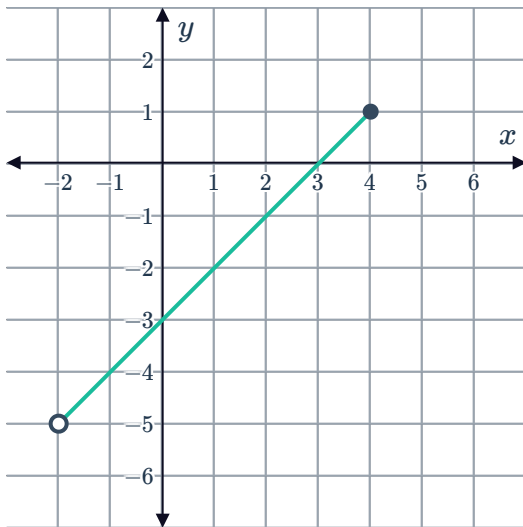
14 For each of the following graphs, state the:

i Domain

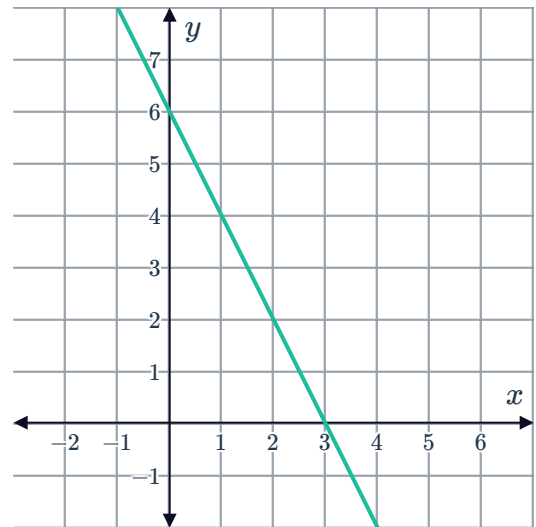
ii Range

iii x - and y -intercepts

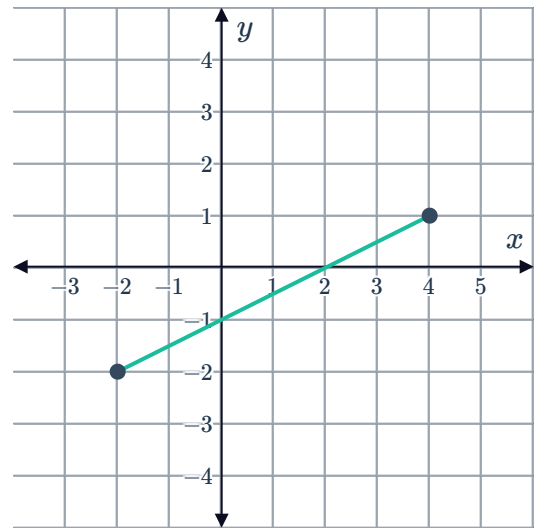
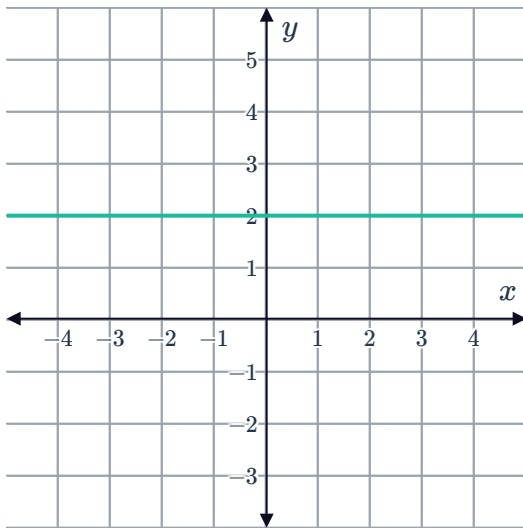
a



b



c



15 The graph shows the height of a soccer ball against the time that has passed after it has been kicked.

- a Find the coordinates of the y -intercept.
- b Interpret the y -value of the y -intercept in this context.
- c Find the coordinates of the x -intercept.
- d Interpret the x -value of x -intercept in this context.

